Customer Segmentation & Retention Alerts

RFM Analysis | Python, Power BI, Make.com

Project report by Aarish Suradkar

1. Project overview

This project segments the customers of a UK online gift retailer using RFM analysis (Recency, Frequency, Monetary value). It then turns the segments into a three-page Power BI dashboard and an automated Make.com workflow that emails the team about high-value customers who have stopped buying.

Headline result: out of 5,878 customers, the Champions segment is 25.2% of customers but generates 69.2% of revenue.

2. Business problem

The retailer treats all customers the same, so marketing spend is spread evenly. The questions this project answers:

- Which customers generate most of the revenue?
- Which valuable customers are drifting away?
- Who should the marketing team contact first, and with what offer?

3. Dataset

- **Source:** Online Retail II (UCI Machine Learning Repository / Kaggle), two CSV files covering Dec 2009 to Dec 2011.
- **Columns used:** Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country.
- **Size:** 1,067,371 raw rows (after combining both files), reduced to 779,425 after cleaning.
- **Customers:** 5,878 unique customers.
- **Total revenue analysed:** about £17.4 million.

4. Tools used

Tool	Purpose
Python (pandas, NumPy, matplotlib)	Cleaning, RFM calculation, scoring, segmentation, exports
Power BI Desktop	Interactive dashboard with DAX measures
Google Sheets	Holds the at-risk customer list
Make.com	Automated alert workflow
VS Code, Git, GitHub	Development and version control

5. Step-by-step process

Step 1: Load and combine the data

The two yearly CSV files (2009-10 and 2010-11) were loaded with pandas and combined into one table. The files carry a hidden BOM character at the start, so they were read with `utf-8-sig` encoding, and the column names were stripped of stray characters.

Step 2: Clean the data

Rows were removed when:

- the Customer ID was missing, since a row cannot be assigned to a customer without it;
- the invoice number started with "C" (cancelled orders);
- quantity or price was zero or negative (returns and bad entries);
- the row was an exact duplicate, which also removes the overlap between the two files.

A `TotalPrice` column (Quantity × Price) was added. The result was 779,425 clean transaction rows.

Step 3: Calculate RFM metrics

A snapshot date was set to one day after the last transaction. For each customer:

- **Recency:** days between the snapshot date and the customer's last purchase.
- **Frequency:** number of unique invoices.
- **Monetary:** total amount spent.

Step 4: Score each metric from 1 to 5

Each metric was divided into quintiles with `pd.qcut`:

- Recency scores were reversed, because fewer days since the last purchase is better.
- Frequency was ranked before scoring, because many customers share the same order count and the bin edges would otherwise be duplicated.

Step 5: Assign segments

Segments were assigned by rules on the Recency and Frequency scores:

Segment	Rule (R score, F score)
Champions	R 4-5 and F 4-5
Loyal Customers	R 3 and F 4-5
Potential Loyalists	R 4-5 and F 3
New Customers	R 4-5 and F 1-2
Need Attention	R 3 and F 1-3
Cant Lose Them	R 1-2 and F 4-5
At Risk	R 1-2 and F 3
Hibernating	R 1-2 and F 2
Lost	all remaining customers

Step 6: Export results

The script exported four CSV files for Power BI: customer-level RFM segments, a segment summary, monthly revenue, and revenue by country. A chart of customers and revenue by segment was also saved.

Step 7: Build the Power BI dashboard

The CSVs were loaded into Power BI and the following DAX measures were created:

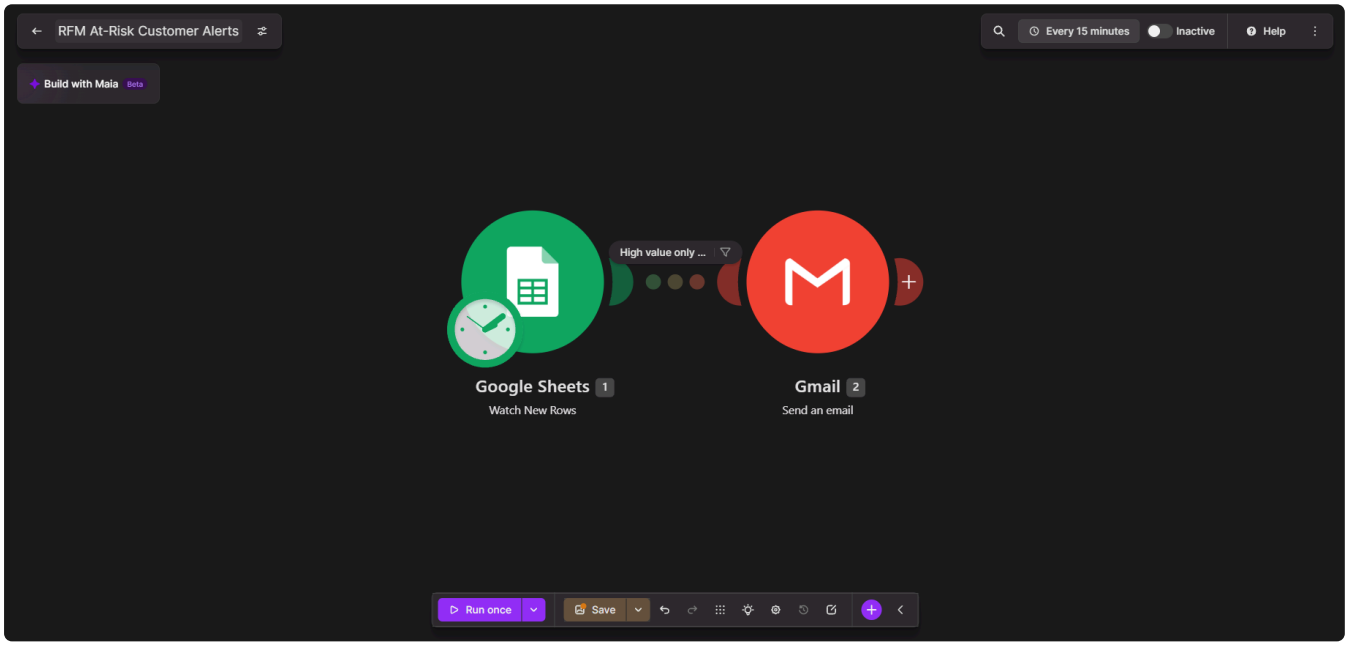
- `Total Revenue`: sum of Monetary
- `Total Customers`: distinct count of Customer ID
- `Avg Revenue per Customer`: revenue divided by customers
- `Revenue %` and `Customer %`: each segment's share of the total
- `Segment Color`: a rule that gives each segment a consistent colour

The dashboard has three pages: Overview, Segments and Action List (described in section 7).

Step 8: Automate the alerts with Make.com

The top 50 customers by spend in the At Risk and Cant Lose Them segments were exported and imported into a Google Sheet. A Make.com scenario then:

1. watches the sheet for new rows (every 15 minutes);
2. filters for customers who have spent more than £1,000;
3. sends an email through Gmail with the customer ID, segment, country, days since last purchase and total spend, plus a suggested action.



6. Results

Segment summary

Segment	Customers	% of customers	Revenue (£)	% of revenue	Avg recency (days)	Avg orders	Avg spend (£)
Champions	1,482	25.2	12,024,330	69.2	20.4	15.7	8,113.58
Loyal Customers	516	8.8	1,691,928	9.7	106.3	8.5	3,278.93
Cant Lose Them	353	6.0	1,090,694	6.3	344.2	7.6	3,089.79
Need Attention	651	11.1	543,149	3.1	109.7	2.1	834.33
At Risk	471	8.0	498,690	2.9	387.8	3.0	1,058.79

Segment	Customers	% of customers	Revenue (£)	% of revenue	Avg recency (days)	Avg orders	Avg spend (£)
Potential Loyalists	439	7.5	479,319	2.8	24.9	3.1	1,091.84
New Customers	443	7.5	392,267	2.3	28.1	1.5	885.48
Hibernating	707	12.0	369,993	2.1	440.2	1.5	523.33
Lost	816	13.9	284,434	1.6	475.8	1.0	348.57
Total	**5,878**	**100**	**17,374,804**	**100**	**201.3**	**6.3**	**2,955.90**

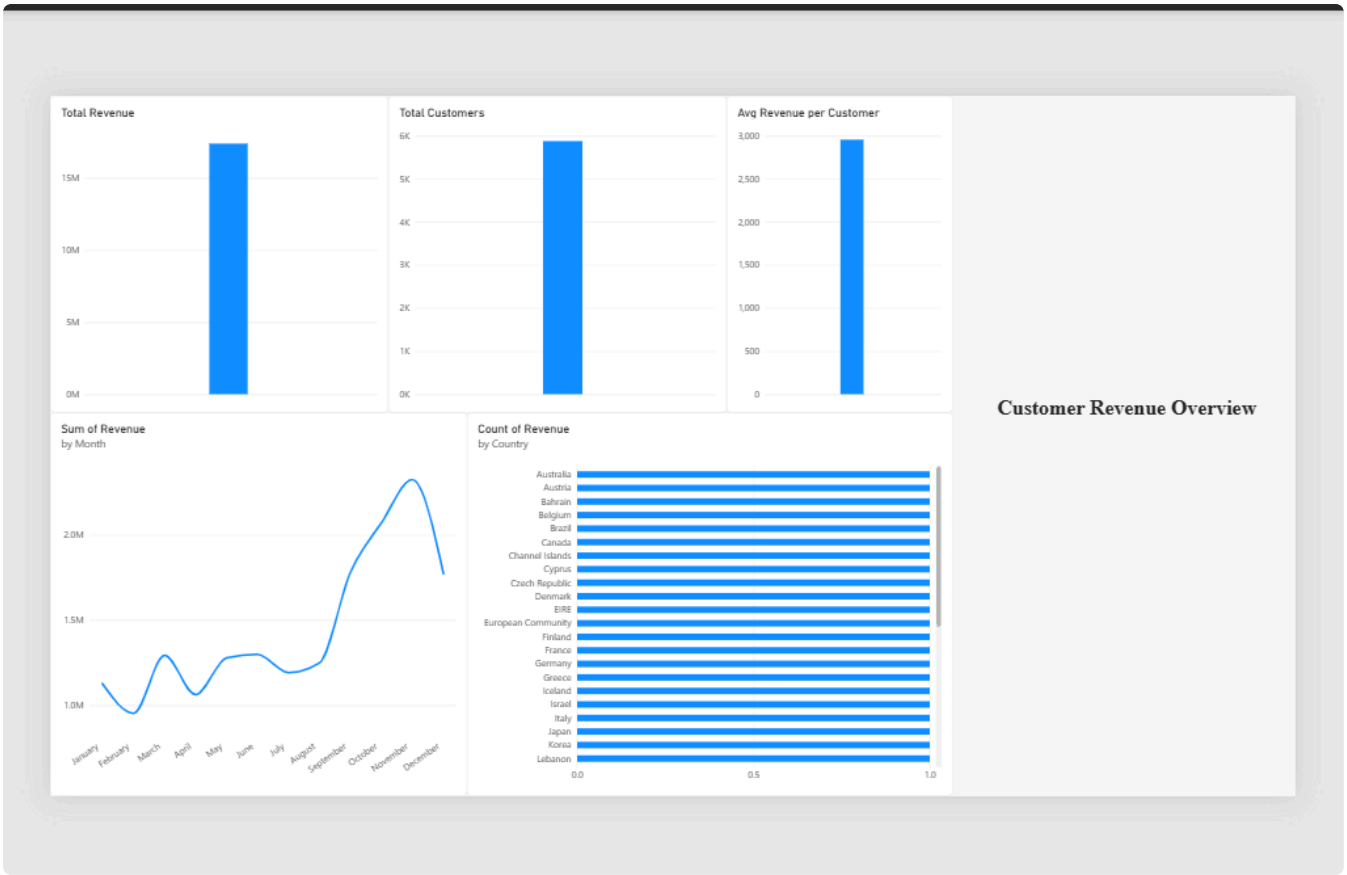
Key findings

1. **Revenue is highly concentrated.** Champions are about a quarter of customers but generate about 69% of revenue. Losing a small number of them would hurt far more than losing many low-value customers.
2. **Cant Lose Them is the best win-back target.** These 353 customers last bought about 344 days ago, but they placed about 7.6 orders and spent about £3,090 each on average. They are lapsed high spenders.
3. **At Risk customers are drifting.** The 471 customers in this group last bought about 388 days ago, so a reactivation campaign is time-sensitive.
4. **Lost and Hibernating customers are low value.** They make up about 26% of customers but only about 3.7% of revenue, and Lost customers average a single order.
5. **Revenue is seasonal.** Monthly revenue rises from September to a peak in November, which suits a gift retailer.

7. Dashboard

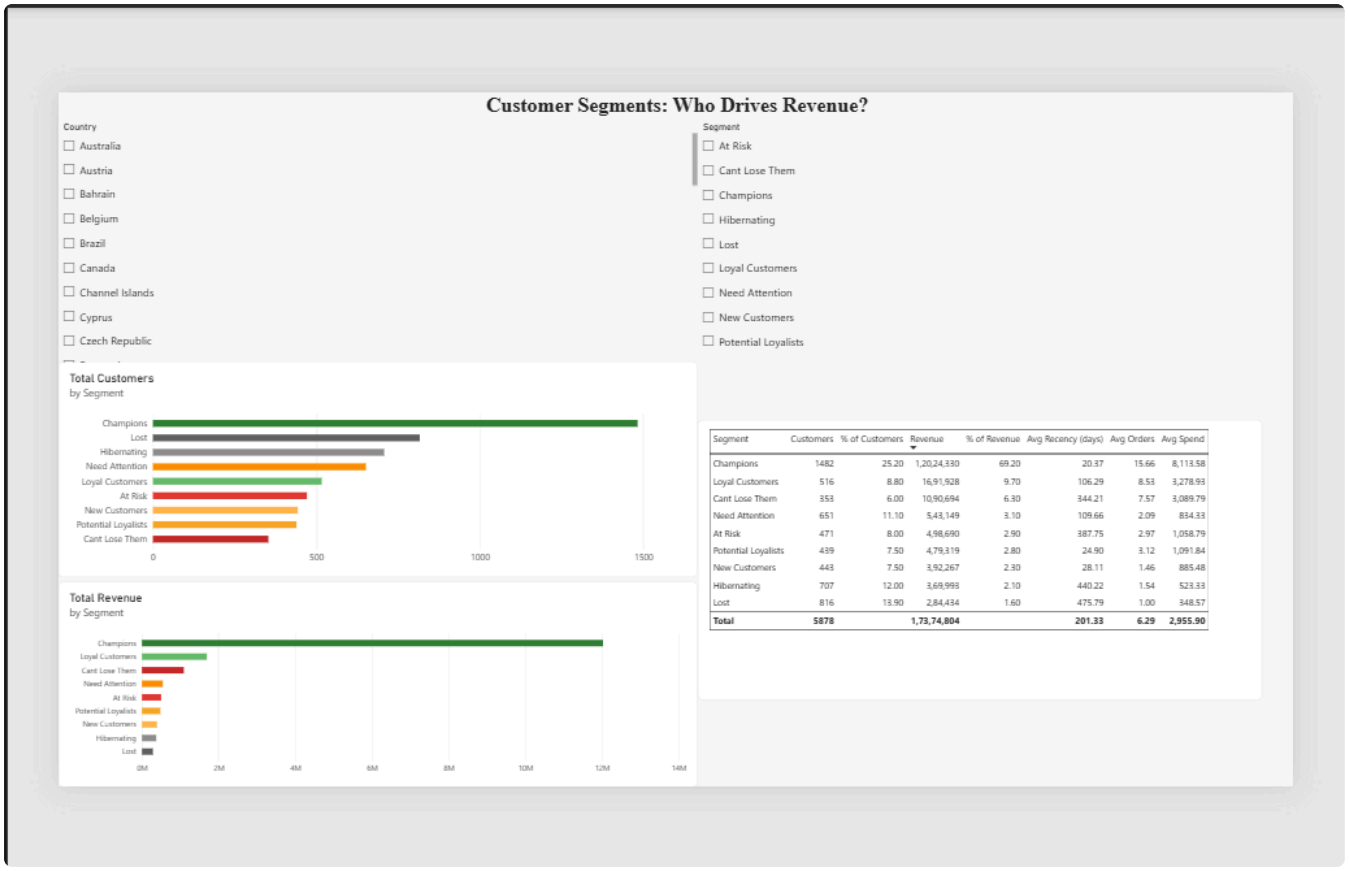
Page 1: Overview

Shows total revenue, total customers and average revenue per customer, the monthly revenue trend and revenue by country.



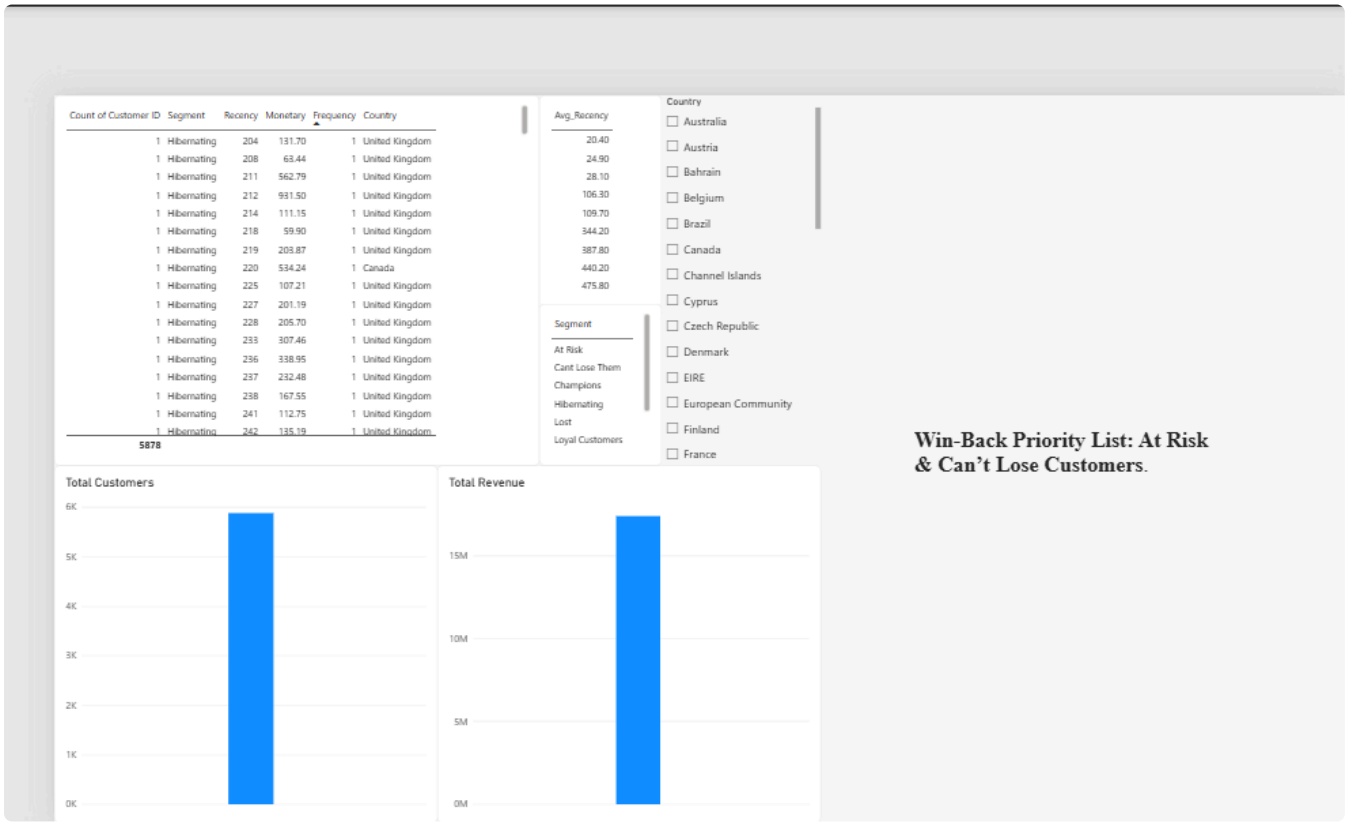
Page 2: Segments

The main analysis page, with customers and revenue per segment, a segment details table, and country and segment filters.



Page 3: Action List

A customer-level list of At Risk and Cant Lose Them customers sorted by past spend, so the marketing team knows exactly who to contact first.



8. Business recommendations

Segment	Recommended action
Champions	Early access to new products, loyalty rewards and a referral programme to protect the main source of revenue
Loyal Customers	Bundle offers and upsell to move them towards Champions
Potential Loyalists	Encourage a regular purchase pattern with follow-up offers
New Customers	Onboarding emails and a second-purchase incentive
Need Attention	Time-limited offers and reminders
Cant Lose Them / At Risk	Personalised win-back offers, starting with the highest spenders (this is the Action List)
Hibernating / Lost	Low-cost reactivation emails only; do not spend heavily on them

9. Challenges and how they were solved

Problem	Cause	Fix
`KeyError: 'Invoice'`	Hidden BOM character made the first column name `ï»¿InvoiceNo`	Read the files with `utf-8-sig` encoding and cleaned the column names
CSV files were not created	`plt.show()` paused the script until the chart window was closed	Replaced it with `plt.close()` so the script runs through to the export
DAX measure errors	Two measures were pasted into one formula, and a measure referred to itself (circular dependency)	Created each measure separately with the correct formula
Dashboard showed 5,350 customers instead of 5,878	A country filter was still active	Cleared the slicer and checked the totals against the Python output

Problem	Cause	Fix
Extra field in a chart title	The colour measure had been dropped into the chart axis	Removed it from the axis and kept it in the colour rule only

10. Limitations and next steps

- Segments are rule-based on R and F scores. K-means clustering could find data-driven groups.
- Customer lifetime value (CLV) could be added to prioritise win-back spend.
- The dataset ends in 2011, so recency is measured from that snapshot date and not from today.
- The Make.com flow is a prototype built on a static sheet. In production it would read from a database or CRM.
- A/B testing the win-back offers would measure whether the campaign works.

11. Skills demonstrated

Data cleaning and transformation (pandas), RFM analysis, customer segmentation, DAX measures, dashboard design in Power BI, workflow automation (Make.com, Google Sheets, Gmail), debugging, Git and GitHub, and turning analysis into business recommendations.